Lesson 3.3

Proving Lines Parallel

Use the angle facts backwards

Run it backwards

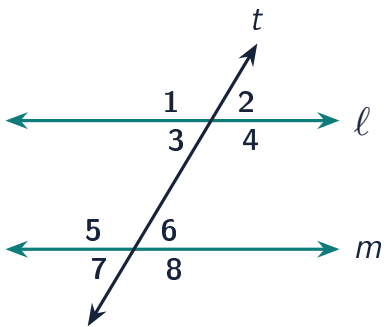
Last lesson: lines parallel $\Rightarrow$ angles congruent.

Run it backwards

Last lesson: lines parallel $\Rightarrow$ angles congruent.

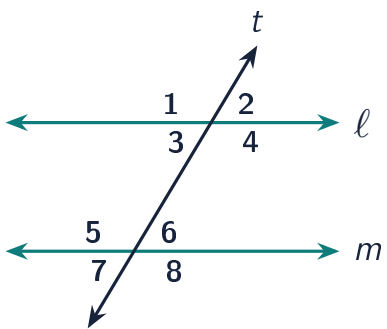
This lesson: angles congruent $\Rightarrow$ **lines parallel**.

The converses



Corresponding angles congruent
$\Rightarrow \ell \parallel m$

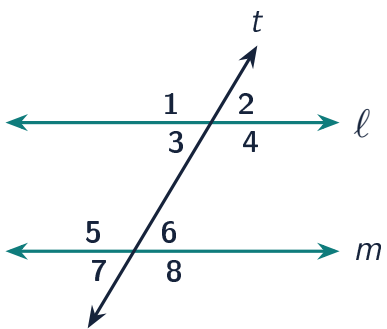
The converses



Corresponding angles congruent $\Rightarrow \ell \parallel m$

Alternate interior or alternate exterior angles congruent $\Rightarrow \ell \parallel m$

The converses

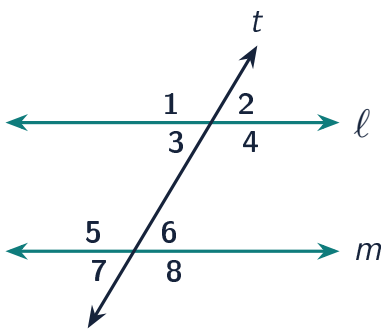


Corresponding angles congruent $\Rightarrow \ell \parallel m$

Alternate interior or alternate exterior angles congruent $\Rightarrow \ell \parallel m$

Same-side interior angles **supplementary** $\Rightarrow \ell \parallel m$

The converses



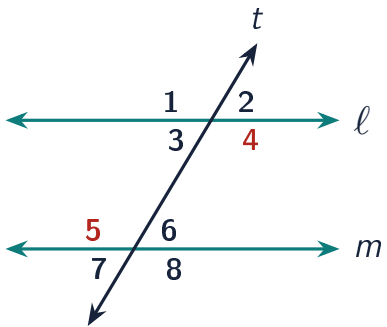
Corresponding angles congruent $\Rightarrow \ell \parallel m$

Alternate interior or alternate exterior angles congruent $\Rightarrow \ell \parallel m$

Same-side interior angles **supplementary** $\Rightarrow \ell \parallel m$

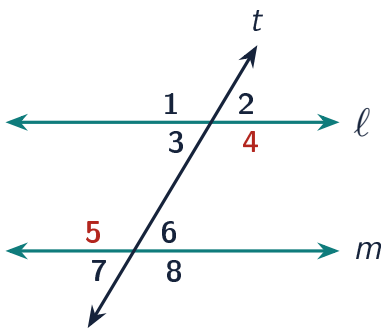
Two lines parallel to the **same** line are parallel.

Prove it



$m\angle 4 = 72°$, $m\angle 5 = 72°$. Is $\ell \parallel m$?

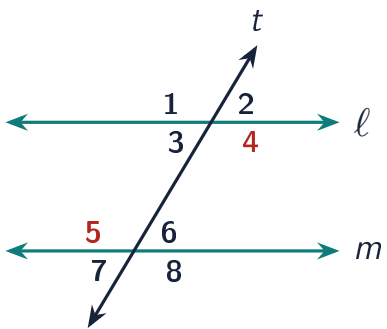
Prove it



$m\angle 4 = 72°$, $m\angle 5 = 72°$. Is $\ell \parallel m$?

$\angle 4$ and $\angle 5$ are alternate interior

Prove it



$m\angle 4 = 72°$, $m\angle 5 = 72°$. Is $\ell \parallel m$?

$\angle 4$ and $\angle 5$ are alternate interior

$\angle 4 \cong \angle 5$, so $\ell \parallel m$ — Alternate Interior Angles Converse

Find x so the lines are parallel

Same-side interior: $(4x + 6)^\circ$ and $(5x - 15)^\circ$

Find x so the lines are parallel

Same-side interior: $(4x + 6)^\circ$ and $(5x - 15)^\circ$

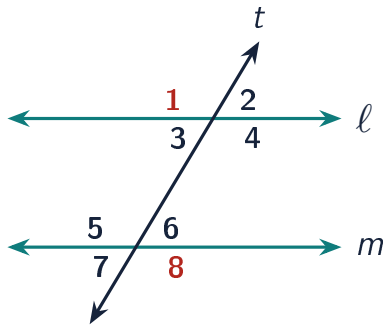
Need a sum of 180°: $9x - 9 = 180 \Rightarrow x = 21$

Watch out

Watch out! Never use "they look parallel." Name the angle pair, show it is congruent (or supplementary), then name the converse.

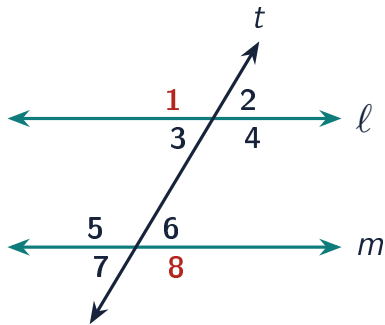
Your turn! Pause the video and try it.

$m\angle 1 = 115°$, $m\angle 8 = 115°$. Is $\ell \parallel m$? Which converse?



Your turn! Pause the video and try it.

$m\angle 1 = 115°$, $m\angle 8 = 115°$. Is $\ell \parallel m$? Which converse?



$\angle 1$, $\angle 8$ are alternate exterior and congruent: yes — Alternate Exterior Angles Converse

Remember

- Congruent corresponding / alternate angles $\Rightarrow$ **parallel**.
- Supplementary same-side interior angles $\Rightarrow$ **parallel**.
- Name the pair, show the angle fact, name the **converse**.

Now practice at your level:

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